

# The University of the South Pacific

Serving the Cook Islands, Fiji, Kiribati, Marshall Islands, Nauru, Niue, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, and Vanuatu.

## School of Computing, Information and Mathematical Sciences

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### CS324: Distributed System

**Learning Outcome:** Outline the importance of distributed systems over stand-alone systems

### Test 1 – Semester 2, 2020

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*Time Allowed: 50 mins*

*Total Marks: 35 (10% of final grade)*

Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

#### INSTRUCTIONS:

- All questions are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

1. Explain the characteristics of Distributed Systems. (2 marks)

Resource sharing  
Openness  
Concurrency  
Scalability  
Fault Tolerance  
Transparency

2. Discuss the importance of architectural styles in Distributed Systems. (2 marks)

ANS: a style is formulated in terms of components, the way that components are connected to each other, the data exchanged between components, and finally how these elements are jointly configured into a system.

3. Explain the relation between architectural style, middleware design pattern and system architecture. (3 marks)

A style is formulated in terms of components, the way that components are connected to each other, the data exchanged between components, and finally how these elements are jointly configured into a system.

Design patterns focuses on the actual organisation of the middleware.

The actual realization of a distributed system requires that we instantiate and place software components on real machines. There are many different choices that can be made in doing so. The final instantiation of a software architecture is also referred to as a **system architecture**.

4. Explain which architectural style is suitable when processes join or leave the system easily. (2 marks)

ANS: Publish Subscribe

5. Discuss the following architectural styles focusing on its importance to distributed systems. Also provide scenarios where these architectural styles would be suitable. (9 marks)

a. Layered style

The basic idea for the layered style is simple: components are organized in a **layered fashion** where a component at layer  $L_j$  can make a **downcall** to a component at a lower-level layer  $L_i$  (with  $i < j$ ) and generally expects a response.

Layered architecture increases **flexibility**, maintainability, and scalability. In a Layered architecture we separate the user interface from the business logic, and the business logic from the data access logic.

Any example that has this architectural style.

b. Object based and service oriented

In essence, each object corresponds to what we have defined as a component, and these components are connected through a procedure call mechanism. In the case of distributed systems, a procedure call can also take place over a network, that is, the calling object need not be executed on the same machine as the called object.

Object-based architectures are attractive because they provide a natural way of encapsulating data (called an object's **state**) and the operations that can be performed on that data (which are referred to as an object's **methods**) into a single entity.

c. Resource based

As an increasing number of services became available over the Web and the development of distributed systems through service composition became more important, researchers started to rethink the architecture of mostly Web-based distributed systems. One of the problems with service composition is that connecting various components can easily turn into an integration nightmare. As an alternative, one can also view a distributed system as a huge collection of resources that are individually managed by components. Resources may be added or removed by (remote) applications, and likewise can be retrieved or modified. This approach has now been widely adopted for the Web and is known as **Representational State Transfer (REST)**

6. Explain the relation between interface, service and protocol. (3 marks)

Protocol describes the rules that parties will follow in order to exchange information.

**service** allows data to be sent from a destination to one or several targets.

**interface** specifies the functions that can be called remotely.

7. Explain the benefits and problems code migration can bring to the distributed systems. (2 marks)

ANS: Benefit: Performance, flexibility

Problems: Code migration brings along problems related to usage of local resources for which it is required that either resources are migrated as well, new bindings to local resources at the target machine are established, or for which system wide network references are used. Another problem is that code migration requires that we take heterogeneity into account.

8. Discuss two communication models suitable for distributed systems. (3 marks)

Remote Procedure Call - An RPC aims at hiding most of the intricacies of message passing, and is ideal for client-server applications.

Message Oriented Middleware - The low-level communication facilities of computer networks are in many ways not suitable again due to their lack of distribution transparency. An alternative is to use a high-level message-queuing model, in which communication proceeds much the same as in e-mail systems.

9. For each of the following forms of coordination in publish subscribe system, explain which communication model would be suitable.
- a. Referentially and temporally coupled. (1 mark)

Direct

- b. Referentially decoupled and temporally coupled. (1 mark)

Event based.

- c. Referentially coupled and temporally decoupled. (1 mark)

Mailbox

- d. Referentially and temporally decoupled. (1 mark)

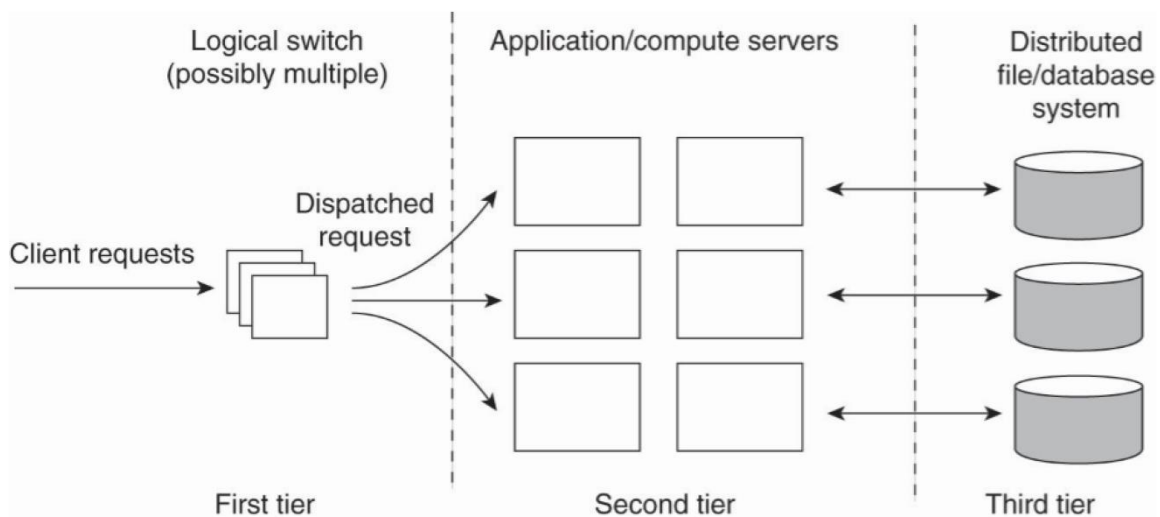
Shared data space.

10. Explain the need for distribution systems. Compare reasons with Standalone systems. (3 marks)

Resource sharing

Scalability

11. Use the following diagram to answer the questions that follow:



- a. How can this architecture help improve the performance of the service? (1 mark)
- Request dispatching in the first tier through multiple switch.
  - Multiple servers can serve multiple clients at one time, even when there are many requests (2<sup>nd</sup> tier)
  - Multiple database servers in third tier.
- b. How can this architecture help improve the failure rate of the service? (1 mark)
- If one server goes down, others are available to serve clients (all tiers).

The End

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